Distributionally Robust Scheduling Problems with Uncertain Release Time and Processing Time

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Abstract

This paper investigates stochastic scheduling problems under uncertain processing time and release time without knowing their exact distribution. Given the mean and variance of both processing time and release time, we present distributionally robust models to minimize the expected total completion time or makespan. We start with single machine scheduling problems and extend our analysis to parallel machine scheduling problems. Since distributionally robust scheduling problems are generally intractable, the linear decision rule is employed. We then develop approximated mixed-integer second-order cone reformulations, which can be solved by state-of-the-art commercial solvers, such as CPLEX and Gurobi. The robustness and effectiveness of our approach are evaluated based on extensive numerical experiments. In contrast to the sample average approximation and the deterministic model's solution, the solution derived from our approach can yield better performance in terms of the 75th, 95th, and 99th quantiles. In addition, the benefit of our model increases with the variability of processing time and release time, as well as the number of jobs. On the other hand, the mean value performance of our approach is very close to the sample average approximation and dominates the deterministic model.

Keywords: Stochastic machine scheduling, Uncertain release time, Uncertain processing time, Distributionally robust optimization

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1. Introduction

Production scheduling is one of the most critical problems in production, manufacturing, and transportation systems. To obtain a good performance measure, such as total completion time and makespan, the decision-maker needs to determine the optimal sequence of the jobs to be processed. Over the last decades, most of the existing literature has studies scheduling problems with specific job parameters, such as deterministic processing time and release time. For a comprehensive and in-depth survey of this field, we refer to the books by Leung (2004) and Pinedo (2008).

Recently, production scheduling with uncertain job parameters has attracted attention from both researchers and practitioners. One typical real-world scheduling problem with uncertain parameters that practitioners need to address is the iron-making process. In this problem, molten iron is first produced by a blast furnace and then is pre-processed sequentially at a pretreatment station. The processing time here is usually short but the release time, which depends on the blast furnace's tapping times, is uncertain (Yue et al. (2018)).

In production systems or the manufacturing industry, uncertainty in processing time and release time is common. Various factors can result in uncertain processing time and release time. For example, uncertainty in processing time can come from machine or tool conditions, worker skill levels, and production environments (Yang and Yu (2002),Xu et al. (2013)), and uncertainty in release time can be affected by arrivals of raw materials or components (Zheng et al. (2019)).

Research on scheduling problems with uncertain job parameters is rising in recent years. Throughout the stochastic scheduling literature, two issues need to be investigated more judiciously. First, most existing literature focuses on addressing uncertainty in processing time. The literature that tries to deal with uncertainty in release time is scarce. As we mentioned, uncertainty in both release time and processing time has been reported in manufacturing systems. Furthermore, Liu et al. (2020b) has shown that the uncertainty in release time may primarily affect the scheduling performance. Therefore, a more reliable study on stochastic scheduling problems should simultaneously account for uncertainty in both release time and processing time.

The description of uncertainty is another pivotal issue in the stochastic scheduling literature. Uncertainty in release time and processing time are described by two main methods in the existing literature. In one stream of the stochastic scheduling literature, the uncertain release time and processing time are regarded as random variables, and the distribution of release time and processing

time is assumed to be known in advance. However, estimating the distribution from historical data is difficult, especially when the size of the historical data is small. Consequently, unknown distribution adds to the difficulty of developing efficient methods to solve stochastic scheduling problems. To address the issue of unknown distributions, a stream of the robust scheduling literature has developed. In robust scheduling problems, typically, the processing time and release time are assumed to reside in an uncertainty set, without any assumptions about their distribution. This assumption is straightforward, but some useful information, such as mean and variance, is overlooked. As a result, the lack of distributional information usually yields an over-conservative solution. Hence, the distributional information, such as mean and variance, needs to be incorporated into stochastic scheduling problems.

Motivated by the above two issues, this paper investigates stochastic scheduling problems, in which: (1) uncertainty in both release time and processing time is considered; (2) the distributional information, i.e., the mean and variance of the processing time and release time, are included, but their exact distribution is unknown in advance. Note that the mean and variance, in contrast to the distribution, can be easily estimated from the historical data. To the best of our knowledge, we are the pioneer to incorporate distributional information about the release time into the scheduling problems. In this paper, we first study single machine scheduling problems and then extend our analysis to parallel machine scheduling problems. The objective is either the expected makespan or the expected total completion time (TCT), which are two of the essential measures in scheduling theory. Efforts to minimize the total completion time tend to reduce the average waiting time of jobs, whereas attempting to minimize the makespan tends to reduce the maximum waiting time of jobs. Hence, these two problems aim to find a job sequence to minimize the makespan or TCT in the presence of uncertain job parameters.

In this paper, distributionally robust optimization (DRO) has been adopted to solve scheduling problems with uncertainties. DRO has been recognized as one of the most powerful approaches to address practical problems affected by uncertainty. As a paradigm for decision-making problems under uncertainty, DRO assumes that the distribution of uncertain parameters belongs to an ambiguity set and reflects that the decision-maker is ambiguity averse. In this paper, the ambiguity set of release time and the processing time is specified by the mean and variance. Scheduling problems under such ambiguity sets are then formulated as two-stage distributionally robust problems. Since

a two-stage distributionally robust problem with an ambiguity set is generally intractable (Bertsimas et al. (2019)), the linear decision rule is employed to provide an approximate formulation. Then, we show that this approximation can be reformulated as a second-order cone program, which can be efficiently solved by state-of-art commercial solvers, such as CPLEX and Gurobi. Extensive numerical experiments are conducted, and the results show that our approach can provide a robust solution to hedge against uncertainty in processing time and release time.

The main contributions are summarized as follows.

- Uncertainty in both processing time and release time has been widely reported, but the research on scheduling problems with uncertain release time is scarce. We are the pioneers to study scheduling problems with uncertain release time processing time and release time to minimize the expected makespan or total completion time.

- Instead of knowing the exact distribution of processing time and release time, the assumption that only mean and variance are known is more realistic. Based on this assumption, we provide distributionally robust optimization programs to model our problems.

- To circumvent the intractability of distributionally robust scheduling problems, we employ the linear decision rule. We then reformulate distributionally robust scheduling models into the second-order cone programs, which can be efficiently solved by commercial solvers, such as CPLEX and Gurobi. We also provide an intuitive understanding of the DRO model.

- Our models can address both single and parallel machine scheduling problems with uncertain release time and processing time. Furthermore, our model can be extended to address a typical scheduling problem with setup costs or resources.

- Extensive numerical experiments show that our models can provide effective solutions and enhance the solutions' robustness under different settings.

The remainder of this paper is organized as follows. After the introduction, we review the literature on stochastic scheduling problems and distributionally robust optimization. In Section 3, we formulate single scheduling problems as a distributionally robust model, in which the ambiguity set is characterized by the mean and variance of release time and processing time. In Section 4, we provide tractable reformulations for the single machine scheduling problems using the linear decision

rule. In Section 5, we conduct extensive numerical experiments and analyze the results under various different parameter settings. The parallel machine scheduling problems are investigated in Section 6. The conclusions are summarized in Section 7.

Notations. Throughout this paper, we use lowercase ***bold*** and uppercase ***bold*** characters to denote vectors and matrices, respectively. Corresponding normal characters denote component-wise elements. Variables with tilde symbols, such as $\tilde{\boldsymbol{r}}$, represent random variables. We use $\boldsymbol{v} \succeq_Q \mathbf{0}$ to indicate that vector $\boldsymbol{v}$ is a second-order cone. $\mathbb{R}_+^n$ denotes an n-dimensional nonnegative real space. $[n]$ represents the set $\{1, 2, 3, \ldots, n\}$ and $[\underline{u}, \overline{u}]^n$ represents the n-dimensional real space with lower bound $\underline{u}$ and upper bound $\overline{u}$. We use $\mathscr{P}_0\left(\mathbb{R}^N\right)$ to represent the set of all probability distributions on $\mathbb{R}^N$. We denote $\mathscr{R}^{N,M}$ as the space of all measurable functions from $\mathbb{R}^N$ to $\mathbb{R}^M$ that are bounded on compact sets.

2. Literature Review

In contrast to deterministic scheduling problems, studies on scheduling problems with uncertain parameters have proliferated in recent years (see, e.g., Drwal (2018), Kasperski (2005), Drwal and Rischke (2016), Chang et al. (2017)). Most of the existing literature assumes that the processing time is uncertain, but research on the scheduling problems with uncertain release time is limited. We review the literature on the scheduling problems with uncertain processing time and scheduling problems with uncertain release time. Since DRO has been adopted to solve our problem, we also review the related DRO literature.

2.1. Scheduling Problems with Uncertain Processing Time

Scheduling problems with uncertain processing time have been studied over the past decades. Suppose that the distribution of processing time is assumed to be known in advance. Pinedo (1983) study a scheduling problem where the processing time is exponentially distributed. To minimize the expected weighted sum of completion time and expected weighted number of late jobs, they establish an efficient and straightforward scheduling policy. Skutella and Uetz (2005) consider a similar problem, and their objective is the expected total weighted completion time. An algorithm based on list scheduling is developed. By assuming that the processing time is uniformly distributed, Alimoradi et al. (2016) investigate a scheduling problem to maximize the probability that the total

flow time does not exceed a predetermined limit. To obtain the optimal sequence, they propose a branch and bound method with a specific branching scheme.

Since the exact distributions of parameters are difficult to obtain in reality, robust optimization (RO) has been applied to solve scheduling problems with uncertainty. RO assumes that the parameters reside in an uncertainty set without a known distribution and aims to provide a robust solution (Ben-Tal et al. (2009)). To the best of our knowledge, Daniels and Kouvelis (1995) is the first to study robust scheduling problems with uncertain processing time. They present exact and heuristic solutions to minimize the expected total completion time. Subsequently, many papers start to investigate robust scheduling problems. Under the assumption that the processing time is uncertain, Xu et al. (2013) study a parallel machine scheduling problem to minimize the maximal regret of makespan. To solve this problem, they derive two exact algorithms by using an iterative relaxation procedure. To minimize the absolute deviation of total flow time from the optimal solution, Lu et al. (2012) reformulated the robust single machine scheduling as a robust constrained shortest path problem, which is solved by a simulated annealing-based algorithmic framework that embeds a generalized label correcting method. Pereira (2016) investigate a similar problem, and their objective is to minimize the maximum absolute regret of the weighted completion time. They present an exact branch-and-bound method to solve the problem.

Since uncertainty is simply characterized through support sets and the distributional knowledge is not considered, it has been observed that RO might propose over-conservative solutions (Wiesemann et al. (2014)). To avoid providing an over-conservative solution, DRO assumes that the distribution of uncertain parameters resides in an ambiguity set, which is described by certain statistics, such as the mean and variance. In addition, these statistics can be easily estimated from data. Chang et al. (2017) is the first to introduce DRO into the production scheduling problem. In their problem, they assume that only the mean and variance of processing time are available. The objective of their problem is the worst-case conditional value-at-risk of the total flow time. They reformulate their problem into a second-order cone program, and numerical experiments show that their model could enhance the optimal job sequence's robustness. Thus, this line of research has attracted attention from many scholars. Chang et al. (2019) extend the DRO to parallel machine scheduling problems with moment ambiguity set. Zhang et al. (2018) study the β-robust scheduling problem under the assumption that only mean and covariance and known. The exact algorithms

are proposed to solve their problem. Liu et al. (2020a) study a distributionally robust flow shop scheduling problem, in which the mean and covariance matrix of processing times are known. The objective is to maximize the profit level. They also show that their proposed hierarchical approach is more efficient than sample average approximation (SAA) and robust SAA methods.

2.2. Scheduling Problems with Uncertain Release Time

The literature on the scheduling problems with uncertain release time is limited. Zheng et al. (2019) investigate a single yard crane (machine) scheduling, where retrieval tasks' release times are uncertain with a known probability distribution. A two-stage stochastic programming is proposed, and the SAA approach is used to solve this problem. Liu et al. (2020b) study a parallel machine scheduling problem where both release time and processing time are stochastic. This problem is formulated as a two-stage stochastic programming, and the SAA is proposed to solve this problem. Under the assumption that only the support set of release time is available, Yue et al. (2018) study a single machine scheduling problem (SMSP) to minimize the worst-case maximum waiting time. They formulate this problem as mixed-integer programming and propose a two-stage heuristic to solve this problem. Bachtler et al. (2020) propose the concept of Gamma-robustness and address the robust absolute and robust regret criteria. To the best of our knowledge, we are the pioneer to deal with the DRO scheduling problem with partial distributional information about uncertain processing time and release time.

2.3. Distributionally Robust Optimization

In this paper, DRO has been adopted to solve a two-stage decision-making problem, and the ambiguity set is specified by the moments. Therefore, we mainly review moments-based DRO and the multi-stage DRO problem. DRO is first introduced by Scarf (1958) to deal with an inventory problem. In their model, the ambiguity set is described by the first and second moments. Since this seminar work, moments-based DRO has been developed and became one of the most powerful approaches to deal with real-world applications in engineering and management (Delage and Ye (2010), Mak et al. (2015)).

In real-world applications, a decision-making problem may not be a single-stage problem. Ben-Tal et al. (2004) is the first to investigate robust multistage decision problems. In the following,

distributionally robust multistage decision problems have been investigated in recent years. Unfortunately, this type of problem is, in general, computationally intractable. To circumvent the intractability, a tractable and efficient approach is the linear decision rule, which restricts adjustable decisions to be affinely dependent on the uncertain parameters. The linear decision rule has been widely used in RO, but the application of linear decision rule in DRO is limited. Bertsimas et al. (2019) extend the linear decision rule by incorporating the auxiliary random variable. In contrast to the traditional linear decision rule, this extension can obtain better performance. Recently, Chen et al. (2020)) propose a new DRO framework called robust stochastic optimization, which unifies a class of scenario-tree-based stochastic linear optimization and many existing DRO models. They also develop a modeling package to help people to implement the multi-stage DRO models. DRO with linear decision rule has been used in several different fields, such as pricing problem (Perakis et al. (2018)), and surgery block allocation (Wang et al. (2019)).

3. Problem Formulation

Production scheduling is one of the crucial problems in production planning decisions. To achieve a good performance, the decision-maker has to allocate jobs to the machines and determine the order of the jobs. For the sake of easy demonstration, we first study the single machine scheduling problems and then investigate the parallel machine scheduling problems in Section 6. In this section, we begin with the formulation of the deterministic model and then provide our distributionally robust scheduling optimization model based on the ambiguity set, which is described by the first two moments of the release time and processing time.

3.1. Deterministic Single Machine Scheduling Problems

Consider a single machine scheduling problem in which a set $\mathscr{J} = \{1, 2, \ldots, n\}$ of jobs are processed on one machine. The processing time of job j is denoted by $p_j, j \in \{1, \ldots, n\}$, which is ready to process at its release time r_j. For simplicity, we denote the processing time vector and release time vector by $\boldsymbol{p} = \{p_1, \ldots, p_n\}$ and $\boldsymbol{r} = \{r_1, \ldots, r_n\}$, respectively. In this paper, we are only concerned with the non-preemptive scheduling discipline, indicating that, once the machine starts to execute a job, it must finish it before executing another. Therefore, the set of feasible

sequences $\mathscr{X}$ can be expressed as follows:

$$\mathscr{X} = \left\{ \boldsymbol{X} \;\middle|\; \begin{array}{l} \sum_{i=1}^{n} x_{ij} = 1, \forall j \in [n] \\ \sum_{j=1}^{n} x_{ij} = 1, \forall i \in [n] \\ x_{ij} \in \{0,1\}, \forall i,j \in [n] \end{array} \right\}$$

In this expression, $x_{ij} = 1$ indicates that job j is processed at the i-th position of the sequence, and $x_{ij} = 0$ otherwise. The first constraint in the above set requires that a given job can only be processed at one position; the second constraint requires that one position should be occupied by only one job, and the third constraint means that the job cannot be divided.

Given a feasible sequence $\boldsymbol{X}$, we denote the start time of the job at the i-th position by t_i. Then, the completion time of the job at the i-th position is $c_i = t_i + \sum_{j=1}^{n} x_{ij} p_j$. Therefore, the makespan (schedule length) can be expressed as $C_{max}(\boldsymbol{X}, \boldsymbol{p}, \boldsymbol{r}) = t_n + \sum_{j=1}^{n} x_{nj} p_j$, and the total completion time can be expressed as $T(\boldsymbol{X}, \boldsymbol{p}, \boldsymbol{r}) = \sum_{i=1}^{n} \left[t_i + \sum_{j=1}^{n} x_{ij} p_j \right]$. In this paper, our goal is to identify a sequence such that the makespan or total completion time is minimized. In general, we denote $f(\boldsymbol{X}, \boldsymbol{p}, \boldsymbol{r})$ by our objective function, which can be the makespan $C_{max}(\boldsymbol{X}, \boldsymbol{p}, \boldsymbol{r})$ or total completion time $T(\boldsymbol{X}, \boldsymbol{p}, \boldsymbol{r})$. Then, our problem can be formulated as a 0-1 integer programming model:

$$Q(\boldsymbol{X}, \boldsymbol{p}, \boldsymbol{r}) = \min_{\boldsymbol{X}, \boldsymbol{t}} \; f(\boldsymbol{X}, \boldsymbol{p}, \boldsymbol{r}) \tag{1}$$

$$\text{s.t } t_i \geq \sum_{j=1}^{n} x_{ij} r_j, \forall i \in [n] \tag{2}$$

$$t_i \geq t_{i-1} + \sum_{j=1}^{n} x_{i-1,j} p_j, \forall i \in \{2, \ldots, n\} \tag{3}$$

$$\boldsymbol{X} \in \mathscr{X} \tag{4}$$

In this formulation, the decision variables are $\boldsymbol{X}$ and $\boldsymbol{t}$, and the objective (1) is the makespan or total completion time. Constraint (2) means that the start time at the i-th position should be greater than or equal to the release time of the job at the i-th position, and constraint (3) indicates that the start time at the i-th position should be greater than or equal to the completion time of the job at the previous position.

3.2. Distributionally Robust Single Machine Scheduling Problem

In practice, the exact value or distribution of the release time and processing time is difficult to be known a priori. The only available information that we can obtain is some statistical information. In the presence of uncertainty in the release time and processing time, the deterministic model is incapable of providing an effective job sequence. Therefore, we model the scheduling problem with uncertain processing time and release time as a DRO model.

To differentiate from the deterministic case, the uncertain processing time and release time are denoted by $\tilde{\boldsymbol{p}} = \{\tilde{p}_1, \ldots, \tilde{p}_n\} \in \mathbb{R}_+^n$ and $\tilde{\boldsymbol{r}} = \{\tilde{r}_1, \ldots, \tilde{r}_n\} \in \mathbb{R}_+^n$, respectively. We assume that the random variable $(\tilde{\boldsymbol{p}}, \tilde{\boldsymbol{r}})$ follows a distribution $\mathbb{P}$, which is unknown in advance, but the mean and variance are available. Define $\tilde{\boldsymbol{z}} = [\tilde{\boldsymbol{p}}, \tilde{\boldsymbol{r}}]^T$, then, we denote the mean vector and standard deviation vector by $\boldsymbol{\mu} = [\boldsymbol{\mu}^p, \boldsymbol{\mu}^r]^T$ and $\boldsymbol{\sigma} = [\boldsymbol{\sigma}^p, \boldsymbol{\sigma}^r]^T$, respectively. Therefore, an ambiguity set that includes all feasible distributions can be expressed as follows:

$$\mathscr{F}_{\mathrm{MM}} = \left\{ \mathbb{P} \in \mathscr{P}_0\left(\mathbb{R}^{2n}\right) \left| \begin{array}{l} \tilde{\mathbf{z}} \sim \mathbb{P} \\ \mathbb{E}_{\mathbb{P}}[\tilde{\mathbf{z}}] = \boldsymbol{\mu} \\ \mathbb{E}_{\mathbb{P}}\left[(\tilde{z}_i - \mu_i)^2\right] \leq \sigma_i^2, \forall i \in [2n] \\ \mathbb{P}\left[\tilde{\mathbf{z}} \in \mathbb{R}_+^{2n}\right] = 1 \end{array} \right. \right\}$$

Under any distribution $\mathbb{P} \in \mathscr{F}_{\mathrm{MM}}$, we require that the expected value of $\tilde{\mathbf{z}}$ equals $\boldsymbol{\mu}$, and the variance of each component of $\tilde{z}_i$ is no greater than σ_i^2, for $i \in [2n]$. Note that $\boldsymbol{\mu}$ and σ_i^2 can be easily estimated from the data.

Now, we are able to provide our distributionally robust model. Given the first-stage decision variable $\boldsymbol{X}$, when the random variable $\tilde{\boldsymbol{z}}$ is realized, the second-stage problem is to determine the start time such as the objective minimum, which can be formulated as follows:

$$Q(\boldsymbol{X}, \tilde{\boldsymbol{z}}) = \min_{\boldsymbol{t}} \ f(\boldsymbol{X}, \tilde{\boldsymbol{z}}) \tag{5}$$

$$\text{s.t } t_i \geq \sum_{j=1}^{n} x_{ij} \tilde{z}_{n+j}, \forall i \in [n] \tag{6}$$

$$t_i \geq t_{i-1} + \sum_{j=1}^{n} x_{i-1,j} \tilde{z}_j, \forall i \in \{2, \ldots, n\} \tag{7}$$

Our aim in the first stage is to identify a sequence that immune all distributions in ambiguity

set $\mathscr{F}_{\mathrm{MM}}$ to minimize the objective. Hence, we formulate our scheduling problem as the following min-max model:

$$\min_{\boldsymbol{X}\in\mathscr{X}} \sup_{\mathbb{P}\in\mathscr{F}_{\mathrm{MM}}} \mathbb{E}_{\mathbb{P}}\left[Q(\boldsymbol{X},\tilde{\boldsymbol{z}})\right] \tag{8}$$

Using this model, we attempt to find a schedule that minimizes the maximal expected makespan or total completion time over the distribution $\mathbb{P}\in\mathscr{F}_{\mathrm{MM}}$.

4. Reformulation of Distributionally Robust Single Machine Scheduling Problems

This section provides the reformulation of the distributionally robust single machine scheduling problem, i.e., problem (8). To describe the reformulation process clearly, we focus on reformulating the total completion time problem in Section 4.1 . Then, the makespan is provided in Section 4.2.

4.1. The Total Completion Time Problem

As a two-stage distributionally robust optimization model, model (8) is in general intractable (Ben-Tal et al. (2004)). To circumvent the intractability, we provide an approximated reformulation based on the linear decision rule. The linear decision rule is one of the most powerful tools for solving the two-stage RO and DRO problems, and it has been used in many applications (Lorca et al. (2016), Ang et al. (2012)).

Observe that, for any given sequence $\boldsymbol{X}$, the second-stage decision variable $\boldsymbol{t}$ can be seen as a mapping of the random variable $\tilde{\boldsymbol{z}}$. Hence, we are ready to present the following proposition.

Proposition 1. *Suppose that the objective is the expected total completion time, then model* (8) *can be equally reformulated as the following program,*

$$\Psi_e = \min_{\boldsymbol{X}\in\mathscr{X}} \sup_{\mathbb{P}\in\mathscr{F}_{\mathrm{MM}}} \mathbb{E}_{\mathbb{P}}\left(\sum_{i=1}^{n}\left[t_i(\tilde{\boldsymbol{z}}) + \sum_{j=1}^{n} x_{ij}\tilde{z}_j\right]\right) \tag{9}$$

$$s.t.\ \ t_i(\boldsymbol{z}) \geq \sum_{j=1}^{n} x_{ij} z_{n+j}, \forall \boldsymbol{z}\in\mathbb{R}_+^{2n}, \forall i\in[n] \tag{10}$$

$$t_i(\boldsymbol{z}) \geq t_{i-1}(\boldsymbol{z}) + \sum_{j=1}^{n} x_{i-1,j} z_j, \boldsymbol{z}\in\mathbb{R}_+^{2n}, \forall i\in\{2,\ldots,n\} \tag{11}$$

$$\boldsymbol{t}\in\mathscr{R}^{2n,n}$$

where $\boldsymbol{t}(\boldsymbol{z}) : \mathbb{R}^{2n} \to \mathbb{R}^n$ *is a decision measure function.*

Note that we do not require $\boldsymbol{t}(\boldsymbol{z})$ to satisfy any explicit form, and the model (9) is still intractable. By restricting the form of the decision measure function, we are able to provide an upper-bounded approximation. In this paper, we employ the linear decision rule, which requires that the decision measure function be a linear function of random variables. Specifically, the decision variable $\boldsymbol{t}$ is defined as follows:

$$\boldsymbol{t} \in \mathscr{L}^{2n,n} = \left\{ \boldsymbol{t} \in \mathscr{R}^{2n,n} \;\middle|\; \begin{array}{l} \exists \boldsymbol{t}^0, \boldsymbol{t}^{1i} \in \mathbb{R}^n, \forall i \in [2n] \\ \boldsymbol{t}(\boldsymbol{z}, \boldsymbol{u}) = \boldsymbol{t}^0 + \sum_{i=1}^{2n} \boldsymbol{t}^{1i} z_i \end{array} \right\}$$

Proposition 2. *Model* (9) *is upper bounded by following model,*

$$\Psi_r = \min_{\boldsymbol{X} \in \mathscr{X}} \sup_{\mathbb{P} \in \mathscr{F}_{\mathrm{MM}}} \mathbb{E}_{\mathbb{P}} \left(\sum_{i=1}^{n} \left[t_i(\tilde{\boldsymbol{z}}) + \sum_{j=1}^{n} x_{ij} \tilde{z}_j \right] \right) \tag{12}$$

$$s.t. \;\; (10), (11)$$

$$\boldsymbol{t} \in \mathscr{L}^{2n,n}$$

Proof. The result is obvious since $\mathscr{L}^{2n,n} \subseteq \mathscr{R}^{2n,n}$. □

Unfortunately, this form of linear decision rule may suffer from some drawbacks (Chen et al. (2008), Goh and Sim (2011)). Bertsimas et al. (2019) propose a more powerful and efficient linear decision rule by introducing an epigraphical random variable. Considering the original ambiguity set $\mathscr{F}_{\mathrm{MM}}$, we introduce an epigraphical random variable $\tilde{\boldsymbol{u}}$ for the term $(\tilde{z}_i - \mu_i)^2, i \in \{1, \ldots, 2n\}$, and then we present a new ambiguity set $\mathscr{G}$,

$$\mathscr{G} = \left\{ \mathbb{P} \in \mathscr{P}_0 \left(\mathbb{R}^{2n} \times \mathbb{R}^{2n} \right) \;\middle|\; \begin{array}{l} (\tilde{\mathbf{z}}, \tilde{\mathbf{u}}) \sim \mathbb{P} \\ \mathbb{E}_{\mathbb{P}}[\tilde{\mathbf{z}}] = \boldsymbol{\mu} \\ \mathbb{E}_{\mathbb{P}}\left[\tilde{u}_i\right] \leq \sigma_i^2, \forall i \in [2n] \\ \mathbb{P}\left[(\tilde{\mathbf{z}}, \tilde{\mathbf{u}}) \in \mathscr{W}_{\mathrm{MM}}\right] = 1 \end{array} \right\}$$

and

$$\mathscr{W}_{\mathrm{MM}} = \left\{ (\mathbf{z}, \mathbf{u}) \in \mathbb{R}^{2n} \times \mathbb{R}^{2n} \;\middle|\; \begin{array}{l} \mathbf{z} \geq \mathbf{0} \\ \mathrm{Case}_{[2n]} \end{array} \right\}$$

with $\text{Case}_{[2n]}$: $\quad \sqrt{(z_i-\mu_i)^2+\left(\frac{u_i-1}{2}\right)^2} \leq \frac{u_i+1}{2}, \quad \forall i \in [2n]$

By incorporating the variable $\tilde{\boldsymbol{u}}$, only linear expectation constraints exist in the ambiguity set. This expression helps us to reformulate DRO models easily. We also note that the ambiguity set, $\mathscr{F}_{MM}$, equals the marginal distributions of $\tilde{\boldsymbol{z}}$ under $\mathbb{P}$, for all $\mathbb{P} \in \mathscr{G}$. With the ambiguity set $\mathscr{G}$, we have the following program,

$$\begin{aligned}
\Psi_l = \min_{\boldsymbol{X}\in\mathscr{X}} \sup_{\mathbb{P}\in\mathscr{G}} \mathbb{E}_{\mathbb{P}} & \left(\sum_{i=1}^{n}\left[t_i(\tilde{\boldsymbol{z}},\tilde{\boldsymbol{u}})+\sum_{j=1}^{n} x_{ij}\tilde{z}_j\right]\right) \\
\text{s.t. } & t_i(\boldsymbol{z},\boldsymbol{u}) \geq \sum_{j=1}^{n} x_{ij} z_{n+j}, \forall(\boldsymbol{z},\boldsymbol{u}) \in \mathscr{W}_{\text{MM}}, \forall i \in [n] \\
& t_i(\boldsymbol{z},\boldsymbol{u}) \geq t_{i-1}(\boldsymbol{z},\boldsymbol{u}) + \sum_{j=1}^{n} x_{i-1,j} z_j, \forall(\boldsymbol{z},\boldsymbol{u}) \in \mathscr{W}_{\text{MM}}, \forall i \in \{2,\ldots,n\} \\
& \boldsymbol{t} \in \mathscr{L}^{4n,n}
\end{aligned} \tag{13}$$

where

$$\mathscr{L}^{4n,n} = \left\{ \boldsymbol{t} \in \mathscr{R}^{4n,n} \;\middle|\; \begin{array}{l} \exists \boldsymbol{t}^0, \boldsymbol{t}^{1i}, \boldsymbol{t}^{2i} \in \mathbb{R}^n, \forall i \in [2n] \\ \boldsymbol{t}(\boldsymbol{z},\boldsymbol{u}) = \boldsymbol{t}^0 + \sum_{i=1}^{2n}\left(\boldsymbol{t}^{1i} z_i + \boldsymbol{t}^{2i} u_i\right) \end{array} \right\}$$

Proposition 3. $\Psi_e \leq \Psi_l \leq \Psi_r$.

Proof. Proposition 3 is a direct result of Theorem 2 in Bertsimas et al. (2019). □

As the decision measure function is a linear function, we are ready to present a tractable reformulation of problem (13), which is stated in the Theorem 1.

Theorem 1. *Model* (13) *is equivalent to model* $[T_{DRO}]$.

$$
\begin{aligned}
[T_{DRO}]\ \inf\ & \boldsymbol{s}^T\boldsymbol{\mu} + \sum_{i=1}^{2n} w_i\sigma_i^2 + \sum_{i=1}^{n} t_i^0 + \sum_{j=1}^{2n}\left(\frac{1}{2}(\alpha_j-\beta_j)-\mu_j\tau_j\right) \\
s.t.\ & t_i^0 \geq \sum_{j=1}^{2n}\left(\frac{1}{2}(a_j^i-b_j^i)-\mu_j c_j^i\right), \quad \forall i\in[n] \\
& t_i^0 - t_{i-1}^0 \geq \sum_{j=1}^{2n}\left(\frac{1}{2}(d_j^{i-1}-e_j^{i-1})-\mu_j f_j^{i-1}\right), \quad \forall i\in\{2,\dots,n\} \\
& [\alpha_j,\beta_j,\tau_j]^T \succeq_Q 0, \quad \forall j\in[2n] && (14)\\
& \tau_j \leq s_j - \sum_{i=1}^{n} t_i^{1j} - \sum_{i=1}^{n} x_{ij}, \quad \forall j\in[n] && (15)\\
& \tau_j \leq s_j - \sum_{i=1}^{n} t_i^{1j}, \quad \forall j\in\{n+1,\dots,2n\} && (16)\\
& \frac{1}{2}(\alpha_j+\beta_j) = w_j - \sum_{i=1}^{n} t_i^{2j}, \quad \forall j\in[2n] && (17)\\
& t_i^{1j} \geq c_j^i, \quad \forall j\in[n], \forall i\in[n] && (18)\\
& t_i^{1j} \geq c_j^i + x_{i,j-n}, \quad \forall j\in\{n+1,\dots,2n\}, \forall i\in[n] && (19)\\
& t_i^{2j} = \frac{1}{2}(a_j^i+b_j^i), \quad \forall j\in[2n], \forall i\in[n] && (20)\\
& t_i^{1j} - t_{i-1}^{1j} \geq f_j^{i-1} + x_{i-1,j}, \quad \forall j\in[n], i\in\{2,\dots,n\} && (21)\\
& t_i^{1j} - t_{i-1}^{1j} \geq f_j^{i-1}, \quad \forall j\in\{n+1,\dots,2n\}, i\in\{2,\dots,n\} && (22)\\
& t_i^{2j} - t_{i-1}^{2j} = \frac{1}{2}(d_j^{i-1}+e_j^{i-1}), \quad \forall j\in[2n], i\in\{2,\dots,n\} && (23)\\
& [a_j^i, b_j^i, c_j^i] \succeq_Q 0, \quad \forall j\in[2n], \forall i\in[n] && (24)\\
& [d_j^{i-1}, e_j^{i-1}, f_j^{i-1}]^T \succeq_Q 0, \quad \forall j\in[2n], i\in\{2,\dots,n\} && (25)\\
& \boldsymbol{w}\geq 0, \boldsymbol{X}\in\mathscr{X}
\end{aligned}
$$

where $\boldsymbol{X}, \boldsymbol{s}, \boldsymbol{w}, \boldsymbol{t}^0, \boldsymbol{t}^1, \boldsymbol{t}^2, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\tau}$, $\boldsymbol{a}^i, \boldsymbol{b}^i, \boldsymbol{c}^i, \forall i\in\{1,\dots,n\}$, *and* $\boldsymbol{d}^{i-1}, \boldsymbol{e}^{i-1}, \boldsymbol{f}^{i-1}, \forall i\in\{2,\dots,n\}$ *are the decision variables.*

Proof. The proof is provided in Appendix. □

Model $[T_{DRO}]$ is a mixed-integer second-order cone program, which can be solved by the state-

of-art commercial solvers, such as CPLEX and Gurobi.

In the following, we provide an clear understanding of the model $[T_{DRO}]$. Let's look at the model $[T_{DRO}]$, we know that the objective function can be rewrited as follows,

$$\begin{aligned}
&\sum_{i=1}^{n} t_i^0 + \sum_{j=1}^{n}\left(\tau_j\mu_j + \sum_{i=1}^{n}(t_i^{1j} + x_{ij})\mu_j\right) + \sum_{j=1+n}^{2n}\left(\tau_j\mu_j + \sum_{i=1}^{n}(t_i^{1j}\right) \\
&+ \sum_{j=1}^{2n}\left(\frac{1}{2}(\alpha_j + \beta_j) + \sum_{i=1}^{n} t_i^{2j}\right)\sigma_j^2 + \sum_{j=1}^{2n}\left(\frac{1}{2}(\alpha_j - \beta_j) - \mu_j\tau_j\right) \\
&\Leftrightarrow \sum_{i=1}^{n}\left[t_i^0 + \sum_{j=1}^{2n}\left(t_i^{1j}\mu_j + t_i^{2j}\sigma_j^2\right) + \mu_i\right] + \sum_{j=1}^{2n}\left[\frac{1}{2}(\sigma_j^2 + 1)\alpha_j + \frac{1}{2}(\sigma_j^2 - 1)\beta_j\right]
\end{aligned}$$

we consider the following problem,

$$\begin{aligned}
\inf \quad & \sum_{j=1}^{2n}\left[\frac{1}{2}(\sigma_j^2 + 1)\alpha_j + \frac{1}{2}(\sigma_j^2 - 1)\beta_j\right] \\
& [\alpha_j, \beta_j, \tau_j]^T \succeq_Q 0, \forall j \in [2n] \\
& \frac{1}{2}(\alpha_j + \beta_j) \geq -\sum_{i=1}^{n} t_i^{2j}, \forall j \in [2n]
\end{aligned}$$

Take dual of this problem, we have

$$\begin{aligned}
\sup \quad & -D^j \sum_{i=1}^{n} t_i^{2j} \\
\text{s.t} \quad & \begin{bmatrix} \frac{1}{2}(\sigma_j^2 + 1 - D^j) \\ \frac{1}{2}(\sigma_j^2 - 1 - D^j) \\ 0 \end{bmatrix} \succeq_Q 0 \Leftrightarrow \sigma_j^2 \geq D^j, \forall j \in [2n] \\
& D^j \geq 0, \forall j \in [2n]
\end{aligned}$$

Therefore, if $\sum_{i=1}^{n} t_i^{2j} \geq 0$, then $D^j = 0$, if $\sum_{i=1}^{n} t_i^{2j} \leq 0$, then $D^j = \sigma_j^2$.

This model can be reformulated as follows,

$$
\begin{aligned}
[T'_{DRO}] \inf \ & \sum_{i=1}^{n} \left[t_i^0 + \sum_{j=1}^{2n} \left(t_i^{1j} \mu_j + t_i^{2j} (\sigma_j^2 - D^j) \right) + \mu_i \right] \\
\text{s.t. } & 0 \le D^j \le \sigma_j^2 && \forall j \in [2n] \\
& t_i^0 \ge \sum_{j=1}^{2n} \left(\frac{1}{2}(a_j^i - b_j^i) - \mu_j c_j^i \right), && \forall i \in [n] \qquad (26) \\
& t_i^0 - t_{i-1}^0 \ge \sum_{j=1}^{2n} \left(\frac{1}{2}(d_j^{i-1} - e_j^{i-1}) - \mu_j f_j^{i-1} \right), && \forall i \in \{2, \ldots, n\} \qquad (27) \\
& (18), (19), (20), (21), (22), (23), (24), (25) \\
& \boldsymbol{X} \in \mathscr{X}
\end{aligned}
$$

In this model, $t_i^0 + \sum_{j=1}^{2n} \left(t_i^{1j} \mu_j + t_i^{2j} (\sigma_j^2 - D^j) \right) + \mu_i$ can be seen as the expected total completion time of the i-th position, which accounts for the effect of both mean and variance of processing time and release time. Note that D^j controls how much proportion of variance should be considered. Although the structure of model $[T'_{DRO}]$ is a bit complicated, it is similar to the structure of deterministic model. In the deterministic model (1), the start time of the i-th position is constrained by two constraints. While in the model $[T'_{DRO}]$, the start time of the i-th position is affected by three parts, and each one is constrained by two constraints. t_i^0 can be regarded as the baseline value, which is affected by two constraints (26) and (27). t_i^{1j} accounts for the effect of mean value, and t_i^{2j} control the impact of the variance. In sum, model $[T'_{DRO}]$ provides an intuitive understanding of the distributionally robust scheduling model, and it also show that the DRO model $[T_{DRO}]$ can capture the effect of mean and variance.

4.2. The Makespan Problem

In this section, we show that the linear decision rule can be used to solve the makespan problem. In fact, the only difference between the total completion time and the makespan problem is the objective function. Under the ambiguity set $\mathscr{G}$, the distributionally robust makespan problem can

be approximately expressed as

$$
\begin{aligned}
[Cmax] \quad \min_{\boldsymbol{X}\in\mathscr{X}} \sup_{\mathbb{P}\in\mathscr{G}} \mathbb{E}_{\mathbb{P}} & \left(t_n(\tilde{\boldsymbol{z}}, \tilde{\boldsymbol{u}}) + \sum_{j=1}^{n} x_{nj}\tilde{z}_j \right) \\
\text{s.t. } & t_i(\boldsymbol{z}, \boldsymbol{u}) \geq \sum_{j=1}^{n} x_{ij} z_{n+j}, \forall (\boldsymbol{z}, \boldsymbol{u}) \in \mathscr{W}_{MM}, \forall i \in [n] \\
& t_i(\boldsymbol{z}, \boldsymbol{u}) \geq t_{i-1}(\boldsymbol{z}, \boldsymbol{u}) + \sum_{j=1}^{n} x_{i-1,j} z_j, \forall (\boldsymbol{z}, \boldsymbol{u}) \in \mathscr{W}_{MM}, \forall i \in \{2, \ldots, n\} \\
& \boldsymbol{t} \in \mathscr{L}^{4n,n}
\end{aligned}
\tag{28}
$$

Using the same technique that we employed for the total completion time, we present the equivalent model. The results are formally stated in Theorem 2.

Theorem 2. $[Cmax]$ *is equivalent to the model* $[Cmax(DRO)]$.

$$
\begin{aligned}
[Cmax(DRO)] \quad \inf \; & \boldsymbol{s}^T \boldsymbol{\mu} + \sum_{i=1}^{2n} w_i \sigma_i^2 + v \\
s.t. \; & v - t_n^0 \geq \sum_{j\in[2n]} \left(\frac{1}{2}(\alpha_j - \beta_j) - \mu_j \tau_j \right) \\
& \tau_j \leq s_j - t_n^{1j} - x_{nj}, j \in [n] \\
& \tau_j \leq s_j - t_n^{1j}, j \in \{n+1, \ldots, 2n\} \\
& \frac{1}{2}(\alpha_j + \beta_j) = w_j - t_n^{2j}, j \in [2n] \\
& [\alpha_j, \beta_j, \tau_j]^T \succeq_Q 0, \forall j \in [2n] \\
& t_i^0 \geq \sum_{j=1}^{2n} \left(\frac{1}{2}(a_j^i - b_j^i) - \mu_j c_j^i \right), \forall i \in [n] \\
& t_i^0 - t_{i-1}^0 \geq \sum_{j=1}^{2n} \left(\frac{1}{2}(d_j^{i-1} - e_j^{i-1}) - \mu_j f_j^{i-1} \right), \forall i \in \{2, \ldots, n\} \\
& \mathbf{w} \geq \mathbf{0}, \mathbf{X} \in \mathscr{X} \\
& (18), (19), (20), (24) \\
& (21), (22), (23), (25)
\end{aligned}
$$

Proof. We omit the proof since it is similar to the proof of Theorem 1. □

4.3. Special Cases

In the above two sections, uncertainty in both release time and the processing time is considered, and the tractable mixed-integer reformulation is presented. In fact, our approach can be used to solve scheduling problems with either uncertain processing time or uncertain release time. To save space, we do not provide the reformulation of these two special problems.

5. Numerical Experiments

In this section, we conduct numerical experiments to validate the robustness and effectiveness of our distributionally robust scheduling models. Due to the structure and results of $[T_{DRO}]$ and $[Cmax(DRO)]$ being similar, we only provide the analysis of $[T_{DRO}]$ here.

In our numerical experiments, we compare DRO model solutions' performance, i.e., $[T_{DRO}]$, with the deterministic and stochastic models' performance. The formulation of deterministic model and stochastic model, i.e., $[T_{DET}]$ and $[T_{SAA}]$ are provided in Section 5.1. First, $[T_{DET}]$, $[T_{SAA}]$ and $[T_{DRO}]$ are solved based on the historical data and the corresponding solutions are denoted by Seq^d, Seq^s, and Seq^{dro}, respectively. Then, we evaluate the performance of Seq^d, Seq^s and Seq^{dro} using the test set. For robustness, our numerical experiments are conducted on different combinations of four parameters, i.e., range of coefficient of variation $\overline{cv}$, the number of jobs n, the number of historical data K, and the type of true distribution of release time and processing time $dist$. Note that $\overline{cv}$ reflects the variability of the processing time and release time. In Section 5.1, we describe the experiment setting and introduce the deterministic model and stochastic model. In Section 5.2, we analyze the performance of $[T_{DRO}]$ solution under different combinations of $(\overline{cv}, n, K, dist)$. Our experiments are conducted on a desktop with a 8-core CPU and 16.00 GB of memory. The overall framework is implemented with the programming language Python 3.7, and the optimization models are solved by Gurobi 9.0.

5.1. Experiment Description

In our experiment, we uniformly draw the true mean processing time $\boldsymbol{\mu}_p$ from $[10, 50]^n$. The true mean release times $\boldsymbol{\mu}_r$ are uniformly generated from $[0, 30n]^n$. The coefficient of variation $\boldsymbol{cv}$ is uniformly chosen from the box set $[0, \overline{cv}]^n$. Then, for job $j, j \in [n]$, the standard deviation of random

processing time σ_{pj} and release time σ_{pj} are $\sigma_{pj} = \boldsymbol{cv}_j \boldsymbol{\mu}_{pj}$ and $\sigma_{rj} = \boldsymbol{cv}_j \boldsymbol{\mu}_{rj}$, respectively. Note that $\boldsymbol{cv}_j$, $\boldsymbol{\mu}_{pj}$ and $\boldsymbol{\mu}_{rj}$ are the j-th component of the vectors. For simplicity, we do not set different $\overline{cv}$ for the release time and processing time. We draw K historical data from a given distribution with true means and variances. Three types of commonly used distributions are considered: the truncated Normal, Lognormal and Gamma distribution. Specifically, we generate K historical data of release time and processing time, i.e., $\{(\hat{\mathbf{p}}^1, \hat{\mathbf{r}}^1), \ldots, (\hat{\mathbf{p}}^K, \hat{\mathbf{r}}^K)\}$. Then, we are able to calculate the sample mean $\hat{\boldsymbol{\mu}}_r, \hat{\boldsymbol{\mu}}_p$ and sample standard deviation $\hat{\boldsymbol{\sigma}}_r, \hat{\boldsymbol{\sigma}}_p$ for the release time and processing time, which are used to solve the distributionally robust model.

Next, we provide the formulation of the deterministic model and stochastic models. The deterministic model $[T_{DET}]$ is formulated as follows,

$$
\begin{aligned}
[T_{DET}] \ \min_{\boldsymbol{X},\boldsymbol{t}} \ & \sum_{i=1}^{n} \left[t_i + \sum_{j=1}^{n} x_{ij} \hat{\boldsymbol{\mu}}_{pj} \right] \\
\text{s.t } & t_i \geq \sum_{j=1}^{n} x_{ij} \hat{\boldsymbol{\mu}}_{rj}, \forall i \in [n] \\
& t_i \geq t_{i-1} + \sum_{j=1}^{n} x_{i-1,j} \hat{\boldsymbol{\mu}}_{pj}, \forall i \in \{2, \ldots, n\}
\end{aligned}
$$

where $\hat{\boldsymbol{\mu}}_{pj}$ and $\hat{\boldsymbol{\mu}}_{rj}$ are the j-th component of the sample mean vector $\hat{\boldsymbol{\mu}}_p$ and $\hat{\boldsymbol{\mu}}_r$, respectively. The solution of $[T_{DET}]$ are denoted by Seq^d.

Stochastic programming is commonly used to solve an optimization problem with uncertainty. Since the distribution of release time and processing time is unknown, SAA is developed to provide an approximation to stochastic programming. We write the SAA formulation as follows.

$$
\begin{aligned}
[T_{SAA}] \ \min_{\hat{\boldsymbol{t}}^k,\boldsymbol{X}} \ & \frac{1}{K} \sum_{k=1}^{K} \sum_{i=1}^{n} \left[\hat{t}_i^k + \sum_{j=1}^{n} x_{ij} \hat{p}_j^k \right] \\
\text{s.t } & \hat{t}_i^k \geq \sum_{j=1}^{n} x_{ij} \hat{r}_j^k, \forall i \in [n], \forall k \in [K] \\
& \hat{t}_i^k \geq \hat{t}_{i-1}^k + \sum_{j=1}^{n} x_{i-1,j} \hat{p}_j^k, \forall i \in \{2, \ldots, n\}, \forall k \in [K] \\
& \boldsymbol{X} \in \mathscr{X}
\end{aligned}
$$

We denote by Seq^s the solution of model $[T_{SAA}]$. Then, we solve the DRO, i.e., $[T_{DRO}]$, based on the sample mean and sample variance and obtain solution Seq^{dro}.

To evaluate the performance of Seq^d, Seq^s and Seq^{dro}, we compute their corresponding total completion time as follows. We randomly draw 10,000 samples from the true distribution as our test set. For each sample, we compute the total completion time corresponding to Seq^d, Seq^s and Seq^{dro}. Then, we calculate the following four measures over the 10,000 samples:

- means, denoted by M_d, M_s and M_{dro}
- q-th percentiles, denoted by PT_d^q, PT_s^q and PT_{dro}^q for $q = 75, 95$ and $q = 99$, respectively

We then use the values of four measures for the deterministic model as our benchmark and compute the gap between the SAA and deterministic models and the gap between the distributionally robust and deterministic models. Specifically, they can be computed by the following expressions:

$$Gap_{saa}^m = \frac{M_d - M_s}{M_d}, Gap_{dro}^m = \frac{M_d - M_{dro}}{M_d}$$
$$Gap_{saa}^q = \frac{PT_d^q - PT_s^q}{PT_d^q}, Gap_{dro}^q = \frac{PT_d^q - PT_{dro}^q}{PT_d^q}, q = 75, 95, 99$$

For each combination of $(\overline{cv}, n, K, dist)$, we generate 100 instances and calculate corresponding results. Then, the average results over 100 instances in each combination are reported.

5.2. Experiment Analysis

In this subsection, we examine the effectiveness and robustness of Seq^{dro} . We mainly focus on the effect of the range of the coefficient variation $\overline{cv}$, number of jobs n and K on the total completion time. We also report the computational time of different approaches.

Effect of $\overline{cv}$: We first analyze the effect of $\overline{cv}$ on the total completion time. Note that a greater $\overline{cv}$ reflects a greater variability in the release time and processing time. Via using different $\overline{cv}$, we are able to examine how variability of release time and processing time affects the performance of the three approaches. In our numerical experiments, the value of $\overline{cv}$ varies from 0.1 to 0.9, and the number of jobs and number of observations are fixed, i.e., $n = 8$ and $K = 30$. Figures 1, 2 and 3 plot the gaps as the $\overline{cv}$ increases under three different distributions.

As shown in these three figures, it is clear that the $[T_{DET}]$ solution is dominated by the $[T_{SAA}]$ and $[T_{DRO}]$ solution under all situations because all of the gaps are greater than 0. A possible

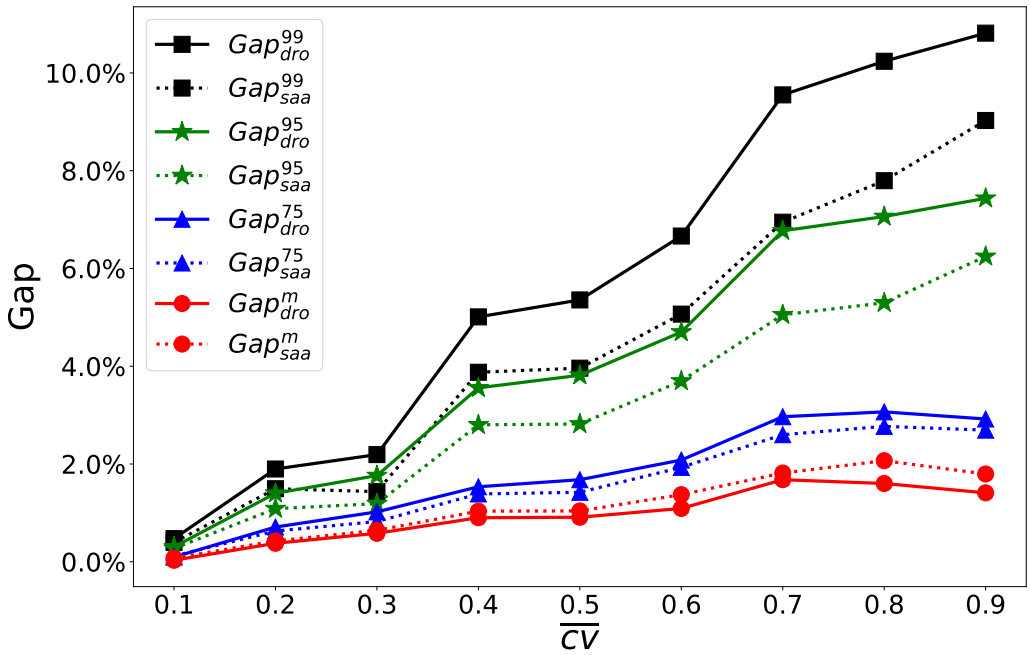

Figure 1: Comparison between SAA and DRO under Lognormal Distribution

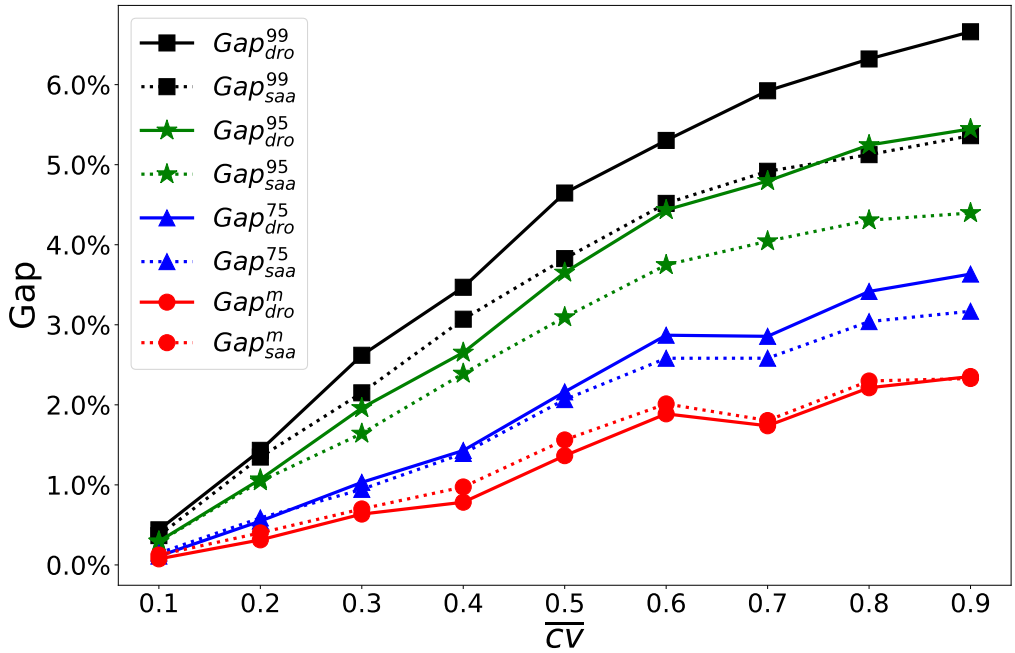

Figure 2: Comparison between SAA and DRO under Normal Distribution

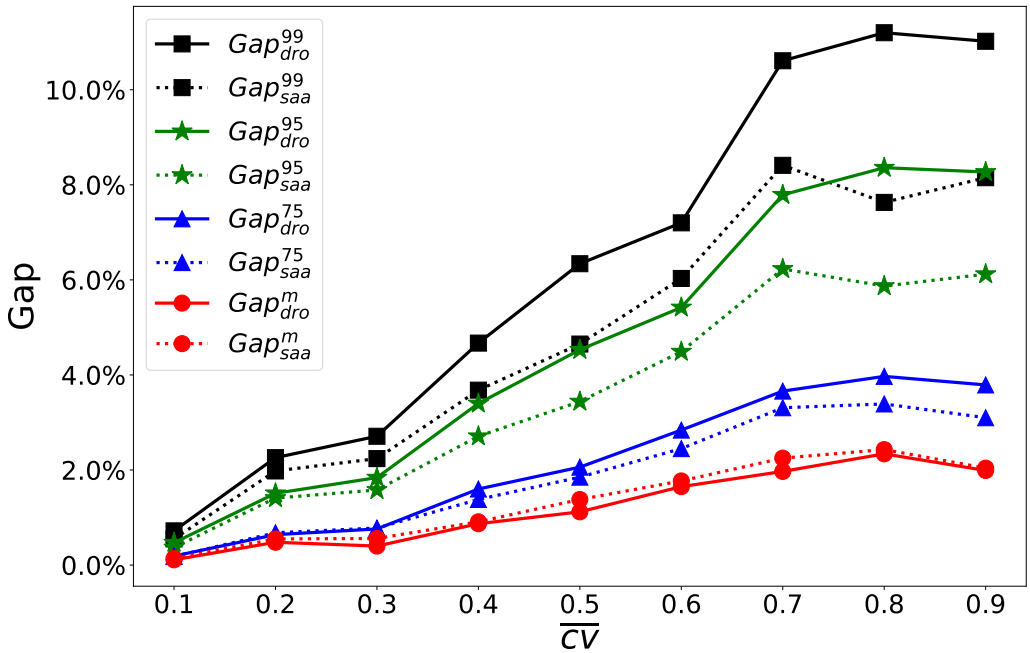

Figure 3: Comparison between SAA and DRO under Gamma Distribution

reason is that the uncertainty does not consider in the deterministic model. Then, Figures 1, 2 and 3 show that $[T_{DRO}]$ solution has a better performance in terms of the q-th percentiles for $q = 75, 95, 99$ than the $[T_{SAA}]$ solution. The better quantile performance implies that $[T_{DRO}]$ is able to avoid the worst situation, and it is remarkable that the advantage of the $[T_{DRO}]$ solution increases as q. Furthermore, the figures also show that the benefit of $[T_{DRO}]$ solution increases as $\overline{cv}$, which suggests that the DRO model can achieve a better performance when then there exists higher variability in the release time and processing time. This finding proves that DRO tends to provide more robust solutions. Finally, we notice that the mean performance of the $[T_{SAA}]$ solution and $[T_{DRO}]$ solution are very close. This phenomenon suggests that the DRO model is a decent approach even for the ambiguity-neutral decision maker.

Effect of the number of jobs Next, we examine the performance of three approaches under different numbers of jobs. In this part, we choose the number of jobs from the set $\{6, 7, 8, 9, 10\}$. The range of the coefficient of variation and number of historical data are fixed, i.e., $\overline{cv} = 0.5$ and $K = 30$. The results are reported in Table 1. Except for the findings that we discussed in the analysis from Figure 1 to Figure 3, we find that the gap between $[T_{SAA}]$ and $[T_{DET}]$ solution and the gap between $[T_{DRO}]$ and $[T_{DET}]$ solution increase as number of jobs grows. For example, the Gap^{95}_{saa}

Table 1: Total Completion Time Results under Different n when $\overline{cv} = 0.5$

dist	n	$[T_{SAA}]$				$[T_{DRO}]$			
		Gap^{m}_{saa}	Gap^{75}_{saa}	Gap^{95}_{saa}	Gap^{99}_{saa}	Gap^{m}_{dro}	Gap^{75}_{dro}	Gap^{95}_{dro}	Gap^{99}_{dro}
Lognormal	6	0.71%	0.92%	1.61%	2.42%	0.55%	1.09%	2.56%	3.90%
	7	0.85%	1.39%	2.78%	3.99%	0.77%	1.61%	3.46%	4.85%
	8	1.38%	1.85%	3.44%	4.65%	1.12%	2.06%	4.53%	6.34%
	9	1.15%	1.78%	3.56%	4.89%	0.96%	2.04%	4.60%	6.40%
	10	1.85%	2.79%	5.25%	6.80%	1.76%	3.18%	6.46%	8.54%
Normal	6	0.94%	1.25%	2.00%	2.52%	0.92%	1.46%	2.48%	3.09%
	7	1.18%	1.55%	2.63%	3.34%	1.29%	2.08%	3.70%	4.69%
	8	1.56%	2.07%	3.10%	3.83%	1.36%	2.16%	3.65%	4.65%
	9	1.71%	2.40%	3.71%	4.49%	1.57%	2.65%	4.56%	5.67%
	10	2.32%	3.18%	4.83%	5.89%	2.22%	3.49%	5.38%	6.55%
Gamma	6	0.65%	0.83%	1.50%	2.02%	0.65%	1.18%	2.38%	3.17%
	7	1.14%	1.62%	2.71%	3.60%	1.06%	1.88%	3.77%	5.03%
	8	1.79%	2.39%	4.64%	6.52%	1.67%	2.48%	5.11%	7.23%
	9	1.65%	2.36%	4.48%	5.98%	1.36%	2.44%	5.04%	6.89%
	10	2.01%	2.92%	5.46%	7.27%	1.86%	3.21%	6.82%	9.36%

under the Lognormal distribution rises from 1.61% to 5.25% as n increases to 10. Meanwhile, the Gap^{95}_{dro} increase from the 2.56% to 6.46%. We also note that the $[T_{DRO}]$ solution achieves a better quantile performance, which confirms the robustness and effectiveness of DRO under different n.

Effect of K This part examines how total completion time changes under a different number of historical data K. Note that the more numerous that the historical data are, the more accurate that estimates are for the mean and variance. Furthermore, $[T_{SAA}]$ can achieve better performance under a large value of K. We choose three reasonable values of K, i.e., $K \in \{10, 30, 50\}$ in our experiments. We also fix other relevant parameters, $\overline{cv} = 0.5, n = 8$. The results are reported in Table 2. For $K = 10$ and $K = 50$, the performance of the three approaches' solution is very similar to the performance of the three approaches' solution with $K = 30$. However, a remarkable finding is that the mean performance of the $[T_{DRO}]$ solution outperforms the $[T_{SAA}]$ solution with $K = 10$. This finding shows that the advantages of the DRO model in the situation of a small historical data size. In manufacturing systems or production industry, small historical data is a common fact.

Computation Time Finally, we report the average computation time of the three approaches in Table 3. On the one hand, we admit that the DRO takes more time to obtain the optimal

Table 2: Total Completion Time Results under Different K when $\overline{cv} = 0.5$ and $n = 8$

dist	K	$[T_{SAA}]$				$[T_{DRO}]$			
		Gap^m_{saa}	Gap^{75}_{saa}	Gap^{95}_{saa}	Gap^{99}_{saa}	Gap^m_{dro}	Gap^{75}_{dro}	Gap^{95}_{dro}	Gap^{99}_{dro}
Lognormal	10	0.96%	1.44%	2.97%	4.23%	1.02%	1.83%	3.79%	5.22%
	30	1.38%	1.85%	3.44%	4.65%	1.12%	2.06%	4.53%	6.34%
	50	1.15%	1.84%	3.68%	4.92%	0.91%	2.01%	4.73%	6.55%
Normal	10	2.09%	2.92%	4.74%	5.83%	2.24%	3.31%	5.18%	6.26%
	30	1.56%	2.07%	3.10%	3.83%	1.36%	2.16%	3.65%	4.65%
	50	1.74%	2.27%	3.50%	4.40%	1.51%	2.39%	4.10%	5.20%
Gamma	10	1.18%	1.69%	3.16%	4.37%	1.28%	2.06%	3.72%	4.99%
	30	1.60%	2.31%	3.82%	5.00%	1.40%	2.50%	4.65%	6.17%
	50	1.43%	1.92%	3.46%	4.57%	1.30%	2.18%	4.27%	5.69%

solution, but on the other hand, the computation time of $[T_{DRO}]$ is still very short. The maximum computation time of $[T_{DRO}]$ is 219.46 seconds. We also find that the computation time of $[T_{DRO}]$ grows as $\overline{cv}$ increases, indicating that the DRO model needs to take more time to search for optimal solutions under greater variability of the release time and processing time.

6. Parallel Machine Scheduling Problems

In fact, our approach can be used to solve parallel machine scheduling problems. Suppose that there are m available machines and n jobs that need to be processed. As usual, the processing time and release time are denoted by $\tilde{\boldsymbol{p}}$ and $\tilde{\boldsymbol{r}}$, respectively. Then, the set of the feasible sequences on m machines can be expressed as follows,

$$\mathscr{X} = \left\{ \boldsymbol{X} \;\middle|\; \begin{array}{l} \sum_{k=1}^{m} \sum_{i=1}^{n} x_{ij}^k = 1, \forall j \in \{1, \ldots, n\} \\ \sum_{j=1}^{n} x_{ij}^k \leq 1, \forall i \in \{1, \ldots, n\}, \forall k = \{1, \ldots, m\} \\ x_{ij}^k \in \{0, 1\}, \forall i, j \in \{1, \ldots, n\}, \forall k = \{1, \ldots, m\} \end{array} \right\}$$

where $x_{ij}^k = 1$ stands for job j is processed at the i-th position of the sequence on the machine k, and $x_{ij}^k = 0$ otherwise. The first constraint in the above set forces that only one job to be processed at one position; the second constraint requires that one position should not be occupied by more than one job. Note that we assume that the sequence on each machine has n positions, and not

Table 3: Computational Time under Different n and $\overline{cv}$ when $K = 30$

dist	n	$\overline{cv} = 0.3$			$\overline{cv} = 0.5$			$\overline{cv} = 0.7$		
		$[T_{DET}]$	$[T_{SAA}]$	$[T_{DRO}]$	$[T_{DET}]$	$[T_{SAA}]$	$[T_{DRO}]$	$[T_{DET}]$	$[T_{SAA}]$	$[T_{DRO}]$
Lognormal	6	0.01	0.04	0.48	0.01	0.05	0.67	0.01	0.06	0.78
	7	0.01	0.08	1.86	0.01	0.09	2.48	0.01	0.11	5.21
	8	0.01	0.17	6.02	0.01	0.21	12.70	0.01	0.28	18.25
	9	0.02	0.31	12.07	0.03	0.45	18.57	0.02	0.42	35.62
	10	0.02	0.29	31.81	0.02	0.42	89.23	0.02	0.48	219.46
Normal	6	0.01	0.05	0.55	0.01	0.05	0.65	0.01	0.05	0.72
	7	0.01	0.09	1.63	0.01	0.12	2.70	0.01	0.09	3.58
	8	0.02	0.27	7.75	0.01	0.20	19.06	0.01	0.17	24.35
	9	0.03	0.45	11.09	0.02	0.38	16.37	0.01	0.25	30.47
	10	0.02	0.32	34.61	0.02	0.36	120.22	0.02	0.41	143.20
Gamma	6	0.01	0.04	0.49	0.01	0.04	0.66	0.01	0.05	0.68
	7	0.01	0.09	1.74	0.01	0.13	2.77	0.01	0.15	4.76
	8	0.02	0.24	10.47	0.01	0.21	13.17	0.01	0.27	15.36
	9	0.02	0.25	18.72	0.02	0.38	15.81	0.01	0.27	40.21
	10	0.02	0.31	31.49	0.02	0.37	77.59	0.02	0.49	181.91

all of the positions are occupied by jobs. The third constraint indicates that the job cannot be divided.

Given a feasible sequence $\boldsymbol{X}$, we denote the start time of the job at the i-th position on machine k by t_{ki}. Then, the completion time of the job at this position is $c_{ki} = t_{ki} + \sum_{j=1}^{n} x_{ij}^{k} \tilde{p}_j$. Next, the makespan on machine k is $C_{max}^{k}(\boldsymbol{X}, \tilde{\boldsymbol{z}}) = t_{kn} + \sum_{j=1}^{n} x_{n,j}^{k} \tilde{p}_j$. Therefore, the makespan of m machines is $C_{max}(\boldsymbol{X}, \tilde{\boldsymbol{z}}) = \max_{k \in [m]} \{C_{max}^{k}\}$, and the total completion time is $T(\boldsymbol{X}, \tilde{\boldsymbol{z}}) = \sum_{k=1}^{m} \sum_{i=1}^{n} t_{ki}$. As usual, we denote $f(\boldsymbol{X}, \tilde{\boldsymbol{z}})$ by our objective function, which can be either $C_{max}(\boldsymbol{X}, \tilde{\boldsymbol{z}})$ or $T(\boldsymbol{X}, \tilde{\boldsymbol{z}})$.

Suppose that the ambiguity set is the same as the that in the above section. The distributionally robust parallel machine scheduling problems can be modeled as follows,

$$\min_{\boldsymbol{x}} \sup_{\mathbb{P} \in \mathscr{F}_{\mathrm{MM}}} \mathbb{E}_{\mathbb{P}} \left[Q(\mathbf{X}, \tilde{\mathbf{z}}) \right] \tag{29}$$
$$\text{s.t } \boldsymbol{X} \in \mathscr{X}$$

where

$$
\begin{aligned}
Q(\boldsymbol{X}, \tilde{\boldsymbol{z}}) = \min_{\boldsymbol{t}} \ & f(\boldsymbol{X}, \tilde{\boldsymbol{z}}) \\
\text{s.t } & t_{ki} \geq \sum_{j=1}^{n} x_{ij}^{k} \tilde{z}_{n+j}, \forall i \in \{1, \ldots, n\}, \forall k \in [m] && (30) \\
& t_{ki} \geq t_{k,i-1} + \sum_{j=1}^{n} x_{i-1,j}^{k} \tilde{z}_{j}, \forall i \in \{2, \ldots, n\}, \forall k \in [m] && (31)
\end{aligned}
$$

The constraint (30) means that the start time at the i-th the position on machine k should greater than or equal to the release time of the job at the i-th position, the constraint (31) means that the start time at the i-th position on machine k should greater that or equal to the completion time of the job at the previous position.

Since the structure of model (29) is the same as that of model (8), it can be reformulated as a mixed-integer second-order cone program. To save space, we do not provide the reformulation here.

Recently, scheduling problems with setup resources or costs have attracted much academic attention (Li et al. (2016), Fanjul-Peyro (2020) and Liu et al. (2020b)). The requirement of resources or costs might be added to our model. For example, Liu et al. (2020b) assume that there is a fixed setup cost θ_{ij}^{A} if job j is assigned to machine i. The objective of this paper is to minimize the the sum of setup costs and the expected penalty costs of jobs' tardiness and earliness. Although the objective is different from our model, it is reasonable to add a budget for the setup cost, i.e., $\sum_{k=1}^{m} \sum_{j=1}^{n} \sum_{i=1}^{n} x_{ij}^{k} \theta_{ij}^{A} \leq \bar{b}$. Then, the scheduling problem with setup costs can be expressed as follows:

$$
\begin{aligned}
& \min_{\boldsymbol{X}} \sup_{\mathbb{P} \in \mathscr{F}_{\mathrm{MM}}} \mathbb{E}_{\mathbb{P}} \left[Q(\mathbf{X}, \tilde{\mathbf{z}}) \right] \\
& \text{s.t } \boldsymbol{X} \in \mathscr{X} \\
& \quad \sum_{k=1}^{m} \sum_{j=1}^{n} \sum_{i=1}^{n} x_{ij}^{k} \theta_{ij}^{A} \leq \bar{b}
\end{aligned}
$$

Due to the budget constraint not changing the structure of the inner problem, this model is still can be solved based on our approach.

7. Conclusion

In this paper, we investigate scheduling problems with uncertain release time and processing time. Since only the mean and variance of release time and processing time are available, we propose the distributionally robust scheduling model (8) to optimize the expected total completion time or makespan. To circumvent the intractability, we provide an approximated mixed-integer second-order cone reformulation based on the linear decision rule to solve this problem. Extensive numerical experiments are conducted on the test set to evaluate the performance of our solution. For the quantile performance, the solution derived from our approximated program outperforms the SAA solutions and deterministic model solutions, and the advantage increase as the variability of release time and processing time grows. When it comes to the mean value performance, the solution derived from our approximated program is very close to the SAA solutions. The numerical experiment results confirm that the proposed DRO models are able to provide robust and effective scheduling solutions in the stochastic production environment. We also show that our model can be extended to solve parallel machine scheduling problems.

Future research could be pursued in the following directions. First, although the approaches proposed in this work can provide high-quality solutions, they come at the cost of computation time. Therefore, an interesting direction would be to develop efficient heuristic algorithms, especially in large-scale scheduling problems. Another extension that deserves to be investigated in the future is to consider different criteria, such as maximum lateness or tardiness. Finally, to capture the correlation between release time and processing time in the scheduling problem and propose an efficient solution would also be valuable research directions.

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8. Appendix

8.1. Proofs of Theorem 1

We start our proof by considering the inner problem of model (13), i.e.,

$$\sup_{\mathbb{P}\in\mathscr{G}} \mathbb{E}_{\mathbb{P}}\left(\sum_{i=1}^{n}\left[t_i(\tilde{\boldsymbol{z}},\tilde{\boldsymbol{u}})+\sum_{j=1}^{n}x_{ij}\tilde{z}_j\right]\right)$$

This problem can be rewritten as the following optimization problem:

$$\begin{aligned}
\sup\ & \int_{(\boldsymbol{z},\boldsymbol{u})\in\mathscr{W}_{MM}} \sum_{i=1}^{n}\left(t_i((\boldsymbol{z},\boldsymbol{u}))+\sum_{j=1}^{n}x_{ij}z_j\right)dF((\boldsymbol{z},\boldsymbol{u})) && (32)\\
\text{s.t.}\ & \int_{(\boldsymbol{z},\boldsymbol{u})\in\mathscr{W}_{MM}} \boldsymbol{z}dF((\boldsymbol{z},\boldsymbol{u}))=\boldsymbol{\mu}\\
& \int_{(\boldsymbol{z},\boldsymbol{u})\in\mathscr{W}_{MM}} u_i dF((\boldsymbol{z},\boldsymbol{u}))\le\sigma_i^2,\forall i\in[2n]\\
& \int_{(\boldsymbol{z},\boldsymbol{u})\in\mathscr{W}_{MM}} dF((\boldsymbol{z},\boldsymbol{u}))=1\\
& dF((\boldsymbol{z},\boldsymbol{u}))\ge 0
\end{aligned}$$

where $F((\boldsymbol{z},\boldsymbol{u}))$ is the cumulative function. The dual problem of problem (32) can be expressed as follows:

$$\inf\ \boldsymbol{s}^T\boldsymbol{\mu}+\sum_{i=1}^{2n}w_i\sigma_i^2+v \tag{33}$$

$$\text{s.t.}\ \boldsymbol{s}^T\mathbf{z}+\sum_{i=1}^{2n}w_iu_i+v\ge\sum_{i=1}^{n}\left(t_i(\boldsymbol{z},\boldsymbol{u})+\sum_{j=1}^{n}x_{ij}z_j\right),\forall(\mathbf{z},\mathbf{u})\in\mathscr{W}_{\mathrm{MM}} \tag{34}$$

$$\mathbf{w}\ge\mathbf{0}$$

$$\boldsymbol{t}\in\mathscr{L}^{4n,n}$$

where $\boldsymbol{s},\boldsymbol{w}$ and v are the dual variables corresponding to the first constraint, the second constraint and the third constraint, respectively. Note that the above problem is still an intractable optimization problem, as there is an infinite number of inequality constraints in (34). We now rewrite

constraint (34) by considering the linear formulation of the decision variable $\boldsymbol{t}$:

$$
\begin{aligned}
&\boldsymbol{s}^T\mathbf{z} + \sum_{i=1}^{2n} w_i u_i + v \geq \sum_{i=1}^{n}\left(t_i(\boldsymbol{z},\boldsymbol{u}) + \sum_{j=1}^{n} x_{ij} z_j\right), \forall(\mathbf{z},\mathbf{u}) \in \mathscr{W}_{\mathrm{MM}} \\
\Leftrightarrow &\boldsymbol{s}^T\mathbf{z} + \sum_{i=1}^{2n} w_i u_i + v \geq \sum_{i=1}^{n}\left(t_i^0 + \sum_{j=1}^{2n}(t_i^{1j} z_j + t_i^{2j} u_j) + \sum_{j=1}^{n} x_{ij} z_j\right), \forall(\mathbf{z},\mathbf{u}) \in \mathscr{W}_{\mathrm{MM}} \\
\Leftrightarrow &v - \sum_{i=1}^{n} t_i^0 \geq \sum_{i=1}^{n}\left(\sum_{j=1}^{2n}(t_i^{1j} z_j + t_i^{2j} u_j) + \sum_{j=1}^{n} x_{ij} z_j\right) - \boldsymbol{s}^T\mathbf{z} - \sum_{i=1}^{2n} w_i u_i, \forall(\mathbf{z},\mathbf{u}) \in \mathscr{W}_{\mathrm{MM}} \\
\Leftrightarrow &v - \sum_{i=1}^{n} t_i^0 \geq \max_{(\mathbf{z},\mathbf{u})\in\mathscr{W}_{\mathrm{MM}}}\left\{\sum_{i=1}^{n}\left(\sum_{j=1}^{2n}(t_i^{1j} z_j + t_i^{2j} u_j) + \sum_{j=1}^{n} x_{ij} z_j\right) - \boldsymbol{s}^T\mathbf{z} - \sum_{i=1}^{2n} w_i u_i\right\} \qquad (35)
\end{aligned}
$$

Then, we consider the maximum problem in the expression (35)

$$
\begin{aligned}
&\max_{(\mathbf{z},\mathbf{u})} \left\{\sum_{i=1}^{n}\left(\sum_{j=1}^{2n}(t_i^{1j} z_j + t_i^{2j} u_j) + \sum_{j=1}^{n} x_{ij} z_j\right) - \boldsymbol{s}^T\mathbf{z} - \sum_{i=1}^{2n} w_i u_i\right\} \qquad (36) \\
&\text{s.t } \boldsymbol{z} \geq 0 \\
&\sqrt{(z_i - \mu_i)^2 + \left(\frac{u_i - 1}{2}\right)^2} \leq \frac{u_i + 1}{2}, \quad \forall i \in [2n]
\end{aligned}
$$

Taking dual of this problem, we can obtain the following problem:

$$
\inf \sum_{j=1}^{2n}\left(\frac{1}{2}(\alpha_j - \beta_j) - \mu_j \tau_j\right) \qquad (37)
$$

$$
\text{s.t. } (14), (15), (16), (17), \qquad (38)
$$

where α_j, β_j and $\tau_j, \forall j \in \{1, \ldots, 2n\}$, are the decision variables. Replace (36) with (37) in (35), and incorporate these results into (33), we have the following equivalent model,

$$\inf \ \boldsymbol{s}^T \boldsymbol{\mu} + \sum_{i=1}^{2n} w_i \sigma_i^2 + v \tag{39}$$

$$\text{s.t. } v \geq \sum_{i=1}^{n} t_i^0 + \sum_{j=1}^{2n} \left(\frac{1}{2}(\alpha_j - \beta_j) - \mu_j \tau_j \right) \tag{40}$$

$$(14), (15), (16), (17),$$

$$\mathbf{w} \geq \mathbf{0}$$

where $\boldsymbol{s}, \boldsymbol{w}, v, \boldsymbol{t}, \alpha_j, \beta_j$ and $\tau_j, j \in [2n]$ are decision variables.

Incorporate outer problem, model (13) can be expressed as follows:

$$\inf \ \boldsymbol{s}^T \boldsymbol{\mu} + \sum_{i=1}^{2n} w_i \sigma_i^2 + v \tag{41}$$

$$\text{s.t. } t_i(\boldsymbol{z}, \boldsymbol{u}) \geq \sum_{j=1}^{n} x_{ij} z_{n+j}, \forall (\boldsymbol{z}, \boldsymbol{u}) \in \mathscr{W}_{\text{MM}}, \forall i \in [n] \tag{42}$$

$$t_i(\boldsymbol{z}, \boldsymbol{u}) \geq t_{i-1}(\boldsymbol{z}, \boldsymbol{u}) + \sum_{j=1}^{n} x_{i-1,j} z_j, \forall (\boldsymbol{z}, \boldsymbol{u}) \in \mathscr{W}_{\text{MM}}, \forall i \in \{2, \ldots, n\} \tag{43}$$

$$(40), (14), (15), (16), (17)$$

$$\mathbf{w} \geq \mathbf{0}, \mathbf{X} \in \mathscr{X}$$

$$\boldsymbol{t} \in \mathscr{L}^{4n,n}$$

Note that we still have infinity constraints in (42) and (43). We now prepare to provide their robust counterparts. We first rewrite constraint (42) as follows:

$$t_i(\boldsymbol{z}, \boldsymbol{u}) \geq \sum_{j=1}^{n} x_{ij} z_{n+j}, \forall (\boldsymbol{z}, \boldsymbol{u}) \in \mathscr{W}_{\text{MM}}, \forall i \in [n]$$

$$\Leftrightarrow t_i^0 \geq \max_{(\boldsymbol{z}, \boldsymbol{u}) \in \mathscr{W}_{\text{MM}}} \left\{ \sum_{j=1}^{n} x_{ij} z_{n+j} - \sum_{j=1}^{2n} (t_i^{1j} z_i + t_i^{2j} u_j) \right\}, \forall i \in [n]$$

For the i-th constraint, consider the problem

$$\max_{(\boldsymbol{z},\boldsymbol{u})} \left\{ \sum_{j=1}^{n} x_{ij} z_{n+j} - \sum_{j=1}^{2n} (t_i^{1j} z_i + t_i^{2j} u_j) \right\}$$
$$\text{s.t } \boldsymbol{z} \geq 0$$
$$\sqrt{(z_i - \mu_i)^2 + \left(\frac{u_i - 1}{2}\right)^2} \leq \frac{u_i + 1}{2}, \quad \forall i \in [2n]$$

For a given $i \in [n]$, we can take the dual of above problem and obtain the following optimization problem:

$$\min \sum_{j=1}^{2n} \left(\frac{1}{2}(a_j^i - b_j^i) - \mu_j c_j^i \right) \tag{44}$$
$$\text{s.t } (18), (19), (20), (24), \tag{45}$$

Next, we rewrite the (43) as follows:

$$t_i(\boldsymbol{z}, \boldsymbol{u}) \geq t_{i-1}(\boldsymbol{z}, \boldsymbol{u}) + \sum_{j=1}^{n} x_{i-1,j} z_j, \forall (\boldsymbol{z}, \boldsymbol{u}) \in \mathscr{W}_{\text{MM}}, \forall i \in \{2, \ldots, n\}$$
$$\Leftrightarrow t_i^0 - t_{i-1}^0 \geq \max_{(\boldsymbol{z},\boldsymbol{u}) \in \mathscr{W}_{\text{MM}}} \left\{ \sum_{j=1}^{2n} \left[(t_{i-1}^{1j} - t_i^{1j}) z_j + (t_{i-1}^{2j} - t_i^{2j}) u_j) \right] + \sum_{j=1}^{n} x_{i-1,j} z_j, \forall i \in \{2, \ldots, n\} \right\}$$

For the i-th constraint, consider the problem

$$\max_{(\boldsymbol{z},\boldsymbol{u})} \left\{ \sum_{j=1}^{2n} \left[(t_{i-1}^{1j} - t_i^{1j}) z_j + (t_{i-1}^{2j} - t_i^{2j}) u_j) \right] + \sum_{j=1}^{n} x_{i-1,j} z_j \right\}$$
$$\boldsymbol{z} \geq 0$$
$$\sqrt{(z_j - \mu_j)^2 + \left(\frac{u_j - 1}{2}\right)^2} \leq \frac{u_j + 1}{2}, \quad \forall j \in [2n]$$

For $i \in \{2, \ldots, n\}$, taking the dual, we have the following model,

$$\min \sum_{j=1}^{2n} \left(\frac{1}{2}(d_j^{i-1} - e_j^{i-1}) - \mu_j f_j^{i-1} \right) \tag{46}$$

$$\text{s.t. } (21), (22), (23), (25) \tag{47}$$

Incorporating model (44) and (46) into problem (41), we obtain one of the main results in this paper. We formally demonstrate it in Theorem 1.